BRAC UNIVERSITY

CSE341 - Microprocessors - Summer 2026

Assignment-2

Total Mark: 40

Question 1 [8 marks]

a) State **two** features of **BIU** and **two** features of **EU**. **[1 + 1]**

b) Illustrate using a block diagram, the internal architecture of Intel 8086 and label the individual components.**[3]**

c) Indicate the purpose of temporary registers. Can the purpose of a temporary register be served using general purpose registers? **[1]**

d) Briefly explain how pipelining speeds up processing. **[1]**

e) Briefly describe the scenario when pipelining fails. **[1]**

Question 2 [10 marks]

a) What are the functions of Flag registers(both status and control flags)? How do they help in error detection? **[1.5 + 0.5]**

b) Mention the **cases** when OF is set with a proper example. **[0.25 + 0.25]**

c) Suppose two hexadecimal numbers **FFCD** and **FFXY** are to be added by an Intel 8086 microprocessor.

i) Given PF = 1 & AF = 0, Find the maximum value for X and Y. Based on the derived values of X and Y, compute the values of all the status flags. **[0.5 + 2]**

ii) Given PF = 0 & AF = 0, Find the minimum value for X and Y. Based on the derived values of X and Y, compute the values of all the status flags. **[0.5 + 2]**

iii) Given AF = 1, find the maximum value for X and minimum value for Y. Based on the derived values of X and Y, compute the values of all the status flags. **[0.5 + 2]**

Question 3 [8 marks]

a) Convert the following assembly instructions to their corresponding machine codes. Show all their workings: **[2 + 2]**

i) MOV DH, [SI+0078h]

ii) MOV [BP+SI+1971h], AX

b) Convert the following machine codes to their corresponding assembly instructions. Show all their workings: **[2 + 2]**

i) 8BC1h. *Also find out the alternative machine code for the assembly instruction derived from this machine code.*

ii) 89912420h

Question 4 [14 marks]

Given, CS = 2010h, SS = 4021h, DS = 615Eh, ES = E74H, IP = 5001H, SI = 888H, DI = 7888H, SP = F57EH, BX = 1000H, BP = 78H

```
MOV AX, 0005H
MOV [BX], AX
MOV [2222H], 0F9H
MOV CX, [BX+1222H]
MOV DX, 0080H
IN AL, DX ; Assume input = 0AH
OUT DX, AX
PUSH AX
CALL FOO
POP CX
MOV DX, CX
MOV [DI], DX
MOV BX, BX[6000H]SI
HLT
JMP [SI+0778H]

FOO PROC:
    ADD AX, 0003H
    RET
FOO ENDP
```

a) Identify the addressing mode(s) used by each instruction. **[4]**

b) For every instruction that accesses the memory,calculate the physical address accessed and specify the data read from or written to that memory location. If the physical address cannot be determined from the given information, state **"Cannot be determined"** and briefly justify your answer. **[3]**

c) Illustrate the stack operation(s) performed during program execution. **[3]**

d) Calculate the physical address where `JMP [SI+0778H]` will jump to if executed. **[2]**

e) Determine the values stored in the registers AX, BX, CX, and DX before and after program execution. **[2]**